

THE GRUT WORKING CORPUS — CHARTER

2026-09-06. Owner-authorized structured first draft. This corpus (Books I–X) is a **working scientific document**, not a publication claim. The goal of this pass is breadth, coherence, traceability, and a readable GRUT worldview. Every chapter is subject to chapter-by-chapter audit; nothing here banks, graduates, or amends the register (74 nodes, sha256 beaeb84e8a6f8468, read-only to this corpus).

GRUT Books — Working Edition v1.0 · D. Ryan Grover · 2026-09-07 · release `zenodo-v1.0` · Markdown is the authoritative source; the PDF is generated from it. Drafted from the frozen record by the RAI builder (Claude, Anthropic) under owner direction — this edition is a publishing pass only: formatting, navigation, and metadata; no claim strengthened or weakened for publication.

The owner's standing directive (verbatim, binding on every book)

Do not invent derivations, do not promote hypotheses to predictions, do not reopen closed gates, and do not create new foundational generations. Build the books from the current frozen record and label every epistemic status honestly.

Corollaries: where the record is silent, the book says so explicitly — an absence map is valid content; write **around** gaps, never silently through them. The word **prediction** is reserved for gate-earned results only, and the PREDICTED set is currently **empty**.

Status vocabulary (display labels, used inline at every substantive claim)

Every substantive claim carries a status block immediately after it:

STATUS: DERIVED — one-line justification (source: `file`)

Display labels and their mapping to the register's eight-status ledger:

display label	ledger status(es)	meaning
DERIVED	DERIVED	exhibited and checked from declared inputs, at a declared scope
RECOVERED	RECOVERED	standard physics reproduced on the declared inputs — largely <i>by identity</i> ; compatibility, not correspondence evidence
ASSUMPTION	AXIOM / EMPIRICAL-INPUT / STRUCTURAL-SELECTION	a declared input; qualify which kind, e.g. ASSUMPTION (STRUCTURAL-SELECTION — chosen, not forced)
HYPOTHESIS	SPECULATIVE	a proposed relation not derived and not validated
UNRESOLVED	UNRESOLVED	an open question the record poses but does not settle
PREDICTED	PREDICTED	gate-earned only; currently empty — no claim carries this label

display label	ledger status(es)	meaning
CLOSED (gate outcome)	—	a computed refusal / no-effect / no-go on the record (e.g. the Γ _T closure); cite the instrument
UNMAPPED	—	the record contains no GRUT account of this topic at all
REVERSED	—	an in-house result later overturned by corrected analysis; part of model history, stated on its face

Scope qualifiers are part of honesty: "DERIVED (flat contract scope, $\omega \gg H$)" is a different claim from "DERIVED." Carry the scope.

Supplementary vocabulary (added 2026-09-06 after the first-draft label census, defining usages the books already carry in disciplined form):

- **STATUS: EMPTY (nothing has earned entry; Book IX governs entry)** — the set-status rendering of canonical claim 21, used wherever a book states the PREDICTED set. It grades a *set*, not a claim; it is the only sanctioned use of EMPTY.
- **STATUS: SHOWN (register tier, ...)** — the register's NATIVE tier for exhibited-and-checked-at-declared-scope, quoted where a book reports a register node in its own vocabulary. Maps to **DERIVED (scoped computed record)** in the display set. The full register-native crosswalk: `shown` → DERIVED (scoped); `derived-pending` → UNRESOLVED (anchor-class); `to-derive` → UNRESOLVED (posed, unattempted or unfinished); `assumed` → ASSUMPTION; `postulate` → ASSUMPTION (AXIOM).
- **Instrument-native grades, quoted** — audited instruments (the RAI Final Boss, the Dialectic Chamber) carry their own grading vocabulary. A book may quote such a grade as a status **ONLY** in the form **STATUS: (instrument grade, quoted at recorded strength)** — re-labeling an instrument verdict into the display set would misquote the record. These quoted grades never upgrade into DERIVED/RECOVERED. The same rule covers recorded audit dispositions quoted verbatim (e.g. "pending refutation if the signal consolidates", `SIGNATURE_AUDIT.md`) and fenced scoped vocabularies (the vacuum-cluster scope), each with its source named inside the block.
- **Qualifier rule** — a defined base label may carry a clarifying suffix or parenthetical ("UNMAPPED as a dynamical sector", "REVERSED-adjacent history, uncertified"); the base label governs, the qualifier narrows.
- **DERIVED dependency rule (global)** — every DERIVED claim must name its immediate premises and whether any premise is GRUT-specific, standard, or selected; where a first-draft block carries only its scope, the chapter-by-chapter audit adds the dependency line rather than the claim being read as unconditional.

This corpus is the **internal working edition**: it keeps the program's governance vocabulary (laundering, fences, banking, W-0). A public-facing edition would translate that vocabulary without weakening it; `books/READERS_MAP.md` is the bridge.

Canonical status table (shared claims MUST use these statuses verbatim across books)

1. Influence-functional/Feynman–Vernon form — **RECOVERED** (generic; u1: the form confers no GRUT-specific content)
2. KMS/FDT lock in equilibrium — **ASSUMPTION** (borrowed standard identity, enforced as a hard admission gate; rung2)
3. $T_{dS} = H/2\pi$ — **DERIVED** (within declarations: forced by Hadamard/KMS on the declared background)
4. Single-pole/finite-memory kernel (rung3) — **UNRESOLVED** (anchor-class, derived-pending; pole-vs-cut open; the Tier-4 computation found a CUT, not a pole, at flat scope)
5. Tier-4 TT kernel, $H^1 = 0$ identically — **DERIVED** (flat contract scope, $\omega \gg H$)
6. Spectral law $s = 5$ — **DERIVED** (flat scope; rejects the framework's own registered $s = 3$)
7. Exact dS constant retarded tail $H^2/4\pi$ — **DERIVED** (exact dS; gapped only at conformal coupling)
8. TT-only projector (p_{tt_ansatz}) — **ASSUMPTION** (STRUCTURAL-SELECTION — CHOSEN, unanimous five-angle interrogation; not forced)
9. $x = 0$ scalar-to-tensor choice — **ASSUMPTION** (STRUCTURAL-SELECTION; x_{no_pin} : the action carries a family, pins nothing)
10. Born measure — **ASSUMPTION** (inherited axiom; the improper-mixture objection stands)
11. Pointer-basis selection via the noise kernel — **RECOVERED** (standard decoherence machinery on declared inputs)
12. USL shape signature (Anastopoulos–Hu driven) — **HYPOTHESIS** (proposed phenomenological relation; shape-only; magnitude verdict quiet-or-faint)
13. $w = -1$ flat — **DERIVED** (within the choices $x = 0$ / pure-TT: the sourced cosmology statement); evolving $w(z)$ — **HYPOTHESIS** (requires the inserted, un-sourced $\tau_2 \sim 1/H_0$, priced +2)
14. $\mu = 4/3$ trace-only endpoint — **CLOSED** (self-exclusion: separate-universe consistency + low- ℓ ISW)
15. Kernel-class discriminator (no purely relaxational kernel crosses $w = -1$; only an oscillatory pole pair) — **DERIVED** (class-level; explicitly not GRUT-specific)
16. Γ_T parameter-free value ($6.19e-63 \cdot H_0$ at 100 Hz) — **CLOSED** (computed NO EFFECT; SPEC outcome REFUSE on the observable route; commits 2116251, 41e1af5)
17. Temporal orientation / Past Hypothesis — **ASSUMPTION** (one relative datum: the half-line/KMS alignment; five dressings audited, every closure consumed it)
18. Finite memory time — **ASSUMPTION**, with REVERSED history on its face (the in-house "no memory time" computation was reversed; exact dS free-field theory forces infinite scale-free memory)
19. GR at zero-memory collapse — **RECOVERED** (largely by identity; KERNEL-STANDARD, scoped)
20. Linear-universe response no-go — **DERIVED** (RRT arm: intervention response reducible for all linear dynamics; escaped only by nonlinearity)
21. The PREDICTED set — **EMPTY** (nothing has earned entry; Book IX governs entry)
22. Flavor, strong-CP, neutrino masses, dark matter, baryogenesis — **UNMAPPED**

23. Type $\text{III}_1 \rightarrow \text{II}_1$ via CLPW crossed product — **RECOVERED (borrowed; audited verdict B-INPUT-RELOCATION: the unexplained input moves, it is not discharged)**
24. $H^1 = 0$ four-channel cancellation + even-degree ladder class (Theorem II) — **DERIVED (the one genuinely new structural identity; carries no confirmatory weight for GRUT)**

Formatting conventions

- Every book opens with the working-draft banner: *"WORKING DRAFT — part of the GRUT working corpus; statuses per `books/CORPUS_CHARTER.md`; subject to chapter-by-chapter audit; nothing here banks."*
- Status blocks inline, immediately after the claim they grade — never deferred to a closing disclaimer.
- Each book ends with two sections: **Sources drawn from** (repo paths) and **Gaps in this book** (explicit, itemized).
- Citations point at repo artifacts by path (and external literature by arXiv id where the record already cites it). Prefer committed artifacts; if an uncommitted session record is load-bearing, mark it "(session record, uncommitted)".
- Readable scientific prose carries the narrative; the status blocks carry the honesty. Neither substitutes for the other.

The corpus

book	file	scope
I	<code>BOOK_I_FOUNDATIONS.md</code>	what GRUT is: constitutive-relational worldview, primitives, status hierarchy, scope
II	<code>BOOK_II_CONSTITUTIVE_FRAMEWORK.md</code>	response, memory, kernels, retardation, operators, effective dynamics
III	<code>BOOK_III_QUANTUM_REALITY.md</code>	QM recovery, measurement/decoherence, Born structure, open systems
IV	<code>BOOK_IV_GRAVITY_SPACETIME_RESPONSE.md</code>	gravitational response, limits, recoveries, gravitational decoherence
V	<code>BOOK_V_MEMORY_THERMODYNAMICS_TIME.md</code>	history dependence, dissipation/noise, arrow of time, KMS/FDT
VI	<code>BOOK_VI_COSMOLOGY.md</code>	background, perturbations, Λ CDM comparison, audited survivals
VII	<code>BOOK_VII_MATTER_SM_INTERFACE.md</code>	electroweak, QCD, Higgs, flavor, neutrinos, dark matter — statuses explicit
VIII	<code>BOOK_VIII_EMERGENCE_OBSERVATION_RELATIONAL.md</code>	observers, operational descriptions, relational/algebraic structure

book	file	scope
IX	BOOK_IX_TESTS_PREDICTIONS.md	experimental philosophy, candidate tests, refused routes, future gates
X	BOOK_X_RESEARCH_PROGRAM.md	what is established, what is conjectural, failure boundaries, roadmap